# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **Coordinate Geometry** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The point on the $y$-axis equidistant from $(2, 3)$ and $(6, 5)$ has $y =$ 1

**(a)** 24 **(b)** 12 **(c)** -12 **(d)** 4

**2.** The area of the triangle formed by $(7, 0)$, $(0, 9)$ and the origin, doubled, is: 1

**(a)** $\frac{63}{2}$ **(b)** 126 **(c)** 16 **(d)** 63

**3.** If $P(x, y)$ is equidistant from $(1, 3)$ and $(7, 5)$ and lies on $y = 0$, then $x =$ 1

**(a)** $\frac{32}{3}$ **(b)** $\frac{16}{3}$ **(c)** 4 **(d)** $-\frac{16}{3}$

**4.** The ratio in which $(-3, 2)$ and $(6, 5)$ are joined by the $y$-axis is 3: 1

**(a)** 6 **(b)** 9 **(c)** 5 **(d)** 3

**5.** The side of the square with diagonal endpoints $(0, 0)$ and $(6, 8)$ has squared length: 1

**(a)** 25 **(b)** 200 **(c)** 50 **(d)** 100

**6.** The three points $(0, 0)$, $(8, 0)$ and $(3, 7)$ enclose area: 1

**(a)** 14 **(b)** 42 **(c)** 56 **(d)** 28

**7.** The point on the $y$-axis equidistant from $(1, 2)$ and $(5, 4)$ has $y =$ 1

**(a)** 9 **(b)** 18 **(c)** 3 **(d)** -9

**8.** The area of the triangle formed by $(11, 0)$, $(0, 4)$ and the origin, doubled, is: 1

**(a)** 22 **(b)** 44 **(c)** 88 **(d)** 15

**9.** **Assertion (A):** The distance of the point $(3, 4)$ from the origin is 5 units. 1
**Reason (R):** The distance of a point $(x, y)$ from the origin is $\sqrt{x^2 + y^2}$.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** The mid-point of the segment joining $(2, -3)$ and $(8, 5)$ is $(5, 1)$. 1

**Reason (R):** The mid-point of the segment joining $(x_1, y_1)$ and $(x_2, y_2)$ is $\left(\frac{x_1 - x_2}{2}, \frac{y_1 - y_2}{2}\right)$.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Show that the points $A(0, 0)$, $B(3, 4)$, $C(8, 4)$, $D(5, 0)$ are the vertices of a rhombus. 2

**12.** Find the value of $k$ for which the points $(1, 5)$, $(2, 8)$ and $(4, k)$ are collinear. 2

**13.** The vertices of a triangle are $(0, 0)$, $(6, 0)$, $(0, 8)$. Find the length of the median from the first vertex. 2

**14.** Show that the points $A(1, 2)$, $B(4, 6)$, $C(9, 6)$, $D(6, 2)$ are the vertices of a rhombus. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Show that the points $A(1, 1)$, $B(5, 4)$, $C(10, 4)$, $D(6, 1)$ are the vertices of a rhombus. 3

**16.** The line joining $(0, 0)$ and $(9, 6)$ is trisected by points $P$ and $Q$. Find their coordinates. 3

**17.** Find the values of $y$ for which the distance between $P(8, y)$ and $Q(2, 3)$ is 10 units. 3

**OR**

Show that the points $A(0, 0)$, $B(6, 8)$, $C(16, 8)$, $D(10, 0)$ are the vertices of a rhombus. 3

**18.** The line joining $(3, 3)$ and $(12, 9)$ is trisected by points $P$ and $Q$. Find their coordinates. 3

**OR**

Find the values of $y$ for which the distance between $P(17, y)$ and $Q(5, 1)$ is 13 units. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** On a town map drawn on a grid (1 unit = 100 m), the school is at $S(2, 3)$, the library at $L(8, 11)$ and the hospital at $H(2, 11)$. 5

(i) Find the coordinates of the mid-point of SL.

(ii) Find the distance SL in units.

(iii) Find the distance LH.

(iv) Show that the triangle SHL is right-angled and state the right-angle vertex.

**20.** The vertices of $\triangle ABC$ are $A(1, 2)$, $B(7, 2)$ and $C(4, 8)$. Find the coordinates of the centroid and the area of the triangle. 5

**OR**

Three vertices of a parallelogram $ABCD$ taken in order are $A(1, 5)$, $B(7, 3)$ and $C(10, 9)$. Find the fourth vertex $D$. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**